

COGNIZANT-DIGITAL-NURTURE-5.0

PL/SQL-

EXCERCISE 1:Control Structures-

scenario1:

```

33 BEGIN
134
135     FOR customer_rec IN
136     (
137         SELECT CustomerID,
138                DOB
139         FROM Customers
140     )
141     LOOP
142
143         IF MONTHS_BETWEEN(
144             SYSDATE,
145             customer_rec.DOB
146         ) / 12 > 60
147         THEN
148
149             UPDATE Loans
150             SET InterestRate = InterestRate - 1
151             WHERE CustomerID =
152                 customer_rec.CustomerID;
153
154             DBMS_OUTPUT.PUT_LINE(
155                 'Discount Applied to Customer ID: '
156                 || customer_rec.CustomerID
157             );
158
159         END IF;
160
161     END LOOP;

```

```
SQL> BEGIN
```

```
FOR customer_rec IN
( ...
```

[Show more...](#)

Discount Applied to Customer ID: 1

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.002

Scenario2:

[SQL Worksheet] ✓ => ⚙ L↓ ⚖

```
176
177 BEGIN
178
179     FOR customer_rec IN
180     (
181         SELECT CustomerID,
182                Balance
183         FROM Customers
184     )
185     LOOP
186
187         IF customer_rec.Balance > 10000
188         THEN
189
190             UPDATE Customers
191             SET IsVIP = 'TRUE'
192             WHERE CustomerID =
193                   customer_rec.CustomerID;
194
195             DBMS_OUTPUT.PUT_LINE(
196                 'VIP Status Granted to Customer ID:
197                 || customer_rec.CustomerID
198             );
199
200         END IF;
201
202     END LOOP;
203
204 END;
205 /
```

SQL> BEGIN

```
        FOR customer_rec IN
        (...
```

Show more...

VIP Status Granted to Customer ID: 1

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.015

Scenario3:

```
7      BEGIN
178
179      FOR loan_rec IN
180      (
181          SELECT LoanID,
182                 CustomerID,
183                 EndDate
184          FROM Loans
185      )
186      LOOP
187
188          IF loan_rec.EndDate <= SYSDATE + 30
189          THEN
190
191              DBMS_OUTPUT.PUT_LINE(
192                  'Reminder: Loan '
193                  || loan_rec.LoanID
194                  || ' for Customer '
195                  || loan_rec.CustomerID
196                  || ' is due on '
197                  || TO_CHAR(
198                      loan_rec.EndDate,
199                      'DD-MON-YYYY'
200                  )
201              );
202
203          END IF;
204
205      END LOOP;
206
207      END;
208      /
```

SQL> BEGIN

```
      FOR loan_rec IN
      (...
Show more...
```

Reminder: Loan 1 for Customer 1 is due on 12-JUL-2026

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.009

Exercise2:

```
CREATE OR REPLACE PROCEDURE SafeTransferFunds
(
    p_from_account NUMBER,
    p_to_account   NUMBER,
    p_amount       NUMBER
)
```

```
IS
```

```
    v_balance NUMBER;
```

```
BEGIN
```

```
    SELECT Balance
    INTO   v_balance
    FROM   Accounts
    WHERE  AccountID =
        |   p_from_account;
```

```
    IF v_balance < p_amount THEN
```

```
        RAISE_APPLICATION_ERROR(
            |   -20001,
            |   'Insufficient Funds'
        );
```

```
    END IF;
```

```
    UPDATE Accounts
    SET   Balance = Balance - p_amount
    WHERE AccountID =
        |   p_from_account;
```

```
    UPDATE Accounts
```

```
BEGIN
```

```
    WHERE AccountID =
```

```
    UPDATE Accounts
    SET   Balance = Balance + p_amount
    WHERE AccountID =
        |   p_to_account;
```

```
    COMMIT;
```

```
    DBMS_OUTPUT.PUT_LINE(
        |   'Transfer Successful'
    );
```

```
EXCEPTION
```

```
    WHEN OTHERS THEN
```

```
        ROLLBACK;

        DBMS_OUTPUT.PUT_LINE(
            |   'Error: '
            |   || SQLERRM
        );
```

```
END;
```

```
/
```

```
BEGIN
```

```
    SafeTransferFunds(
```

```
        |   1,
        |   2,
```

```

267 BEGIN
268
269     SafeTransferFunds(
270         1,
271         2,
272         500
273     );
274
275 END;
276 /

```

Query result **Script output** DBMS output Explain



Procedure SAFETRANSFERFUNDS compiled

Elapsed: 00:00:00.020

> BEGIN

```

    SafeTransferFunds(
        1,...
w more...

```

Transfer Successful

SQL procedure successfully completed.

Elapsed: 00:00:00.014

Scenario2:

```

322 BEGIN
323
324     UpdateSalary(
325         1,
326         10
327     );
328
329 END;
330 /

```

Query result **Script output** DBMS output Explain



Elapsed: 00:00:00.028

SQL> BEGIN

```

    UpdateSalary(
        1,...
Show more...

```

Salary Updated Successfully

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.017

```

332 BEGIN
333
334     UpdateSalary(
335         999,
336         10
337     );
338
339 END;
340 /

```

Query result **Script output** DBMS output



Elapsed: 00:00:00.017

SQL> BEGIN

```

        UpdateSalary(
          999,...
Show more...

```

Employee ID Not Found

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.004

Scenario3:

```

392
393 BEGIN
394
395     AddNewCustomer(
396         3,
397         'Michael Scott',
398         TO_DATE(
399             '1980-05-20',
400             'YYYY-MM-DD'
401         ),
402         25000
403     );
404
405 END;
406 /

```

Query result **Script output** DBMS output



SQL> BEGIN

```

        AddNewCustomer(
          3,...
Show more...

```

Customer Added Successfully

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012

```
408 BEGIN
409
410     AddNewCustomer(
411         3,
412         'David',
413         TO_DATE(
414             '1995-01-01',
415             'YYYY-MM-DD'
416         ),
417         5000
418     );
419
420 END;
421 /
```

Query result

Script output

DBMS output

Ex



SQL> BEGIN

```
        AddNewCustomer(
          3,...
Show more...
```

Customer ID Already Exists

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008

Exercise3:

Stored Procedures

Scenario1-

```
408 BEGIN
409
410     AddNewCustomer(
411         3,
412         'David',
413         TO_DATE(
414             '1995-01-01',
415             'YYYY-MM-DD'
416         ),
417         5000
418     );
419
420 END;
421 /
422 CREATE OR REPLACE PROCEDURE ProcessMonthlyInte
423 IS
424 BEGIN
425
426     UPDATE Accounts
427     SET Balance = Balance + (Balance * 0.01)
428     WHERE AccountType = 'Savings';
429
430     COMMIT;
431
432     DBMS_OUTPUT.PUT_LINE(
433         'Monthly Interest Applied Successfully'
434     );
435
436 EXCEPTION
437
438     WHEN OTHERS THEN
439
440         DBMS_OUTPUT.PUT_LINE(
441             'Error: ' || SQLERRM
```

```
444 END;
445 /
446
447 BEGIN
448
449     ProcessMonthlyInterest;
450
451 END;
452 /
```

Query result Script output DBMS output E



Procedure PROCESSMONTHLYINTEREST compiled

Elapsed: 00:00:00.026

SQL> BEGIN

 ProcessMonthlyInterest;

 ...

Show more...

Monthly Interest Applied Successfully

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.020

```

454 SELECT AccountID,
455        AccountType,
456        Balance
457 FROM Accounts;

```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.005 seconds

	ACCOUNTID	ACCOUNTTYPE	BALANCE
	1	Savings	505
	2	Checking	2000

Scenario2: Before Bonus-

```

461 SELECT EmployeeID,
462        Name,
463        Department,
464        Salary
465 FROM Employees;

```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.006 seconds

	EMPLOYEEID	NAME	DEPARTMENT	SALARY
1	2	Ranbir	IT	60000
2	1	Alia	HR	77000

```

467 CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus
468 (
469     p_department VARCHAR2,
470     p_bonus_percentage NUMBER
471 )
472 IS
473 BEGIN
474
475     UPDATE Employees
476     SET Salary = Salary +
477           (Salary * p_bonus_percentage / 100)
478     WHERE Department = p_department;
479
480     COMMIT;
481
482     DBMS_OUTPUT.PUT_LINE(
483         'Bonus Applied Successfully'
484     );
485
486 EXCEPTION
487
488     WHEN OTHERS THEN
489
490         DBMS_OUTPUT.PUT_LINE(
491             'Error: ' || SQLERRM
492         );
493
494 END;
495 /

```

Query result Script output DBMS output Explain Plan SQL history

Download

Procedure UPDATEEMPLOYEEBONUS compiled

```

494 END;
495 /
496
497 BEGIN
498
499     UpdateEmployeeBonus(
500         'IT',
501         10
502     );
503
504 END;
505 /

```

Query result **Script output** DBMS output



SQL> BEGIN

```

        UpdateEmployeeBonus(
        'IT',...

```

[Show more...](#)

Bonus Applied Successfully

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.023

After Bonus:

```

SELECT EmployeeID,
       Name,
       Department,
       Salary
FROM Employees;

```

Result **Script output** DBMS output Explain Plan SQL history

Download Execution time: 0.001 seconds

EMPLOYEEID	NAME	DEPARTMENT	SALARY
2	Ranbir	IT	66000
1	Alia	HR	77000

